CS 540 Module Two Activity Guidelines and Rubric

Overview

Converting raw, unstructured data into a clean, structured format is a core skill used across industries that rely on data-informed decision making. Using Python coding to prepare data is essential for accurate analysis, reporting, and visualization.

Professionals in fields such as healthcare, business, public policy, and education are expected to work confidently with real-world data sets. Structuring data correctly ensures consistency, enables deeper insights, and supports effective communication of findings. The ability to organize messy input into usable formats is a foundational capability for advanced data analysis tasks in both academic and workplace environments.

Scenario

You've been hired as a data analyst on a data consulting team. The data set your team received is in a raw unaltered state that may be in a structured, semi-structured, or unstructured format with potentially messy entries, including erroneous records, null values, missing headers, mixed data types, or inconsistent delimiters. Before the team can run any analysis or generate reports, they need you to clean up the data and organize it into a clear, tabular format. Your job is to use Python to transform the raw file into a structured table that meets professional data standards and is ready for analysis.

Directions

For this activity, you will write Python code to apply basic data wrangling techniques that uncover patterns and relationships within a data set. You will use your development environment to complete this assignment. When you are ready to begin your work, access Codio by clicking the title of the next task. To begin, choose a data set from the Module Two Activity data folder in Codio.

When the assignment is complete, download the HTML version of your project by clicking on **File**, then **Download As**, then **HTML (.html)** from the Jupyter menu bar. Save the file to your local machine.

Note: *if the code block was not run in the Notebook, it may not render in the HTML output of the Notebook. To fix missing blocks, run the code block, then download as HTML.*

Specifically, you must address the following rubric criteria:

Part One: Python Code

1. Examine the **structure** of the provided data set using Python code. Consider the following:
    A. Are there issues with inconsistent formatting?
    B. Are there missing headers?
    C. Are there gaps in the data?

2. Clean the raw data using **Python functions** to remove and standardize messy entries.
3. Convert the cleaned data into a **tabular format**.
4. **Validate your table** for consistent column names, correct data types, and missing values.

Part Two: Reflection

- Explain how transforming raw data into table format **supports analysis and decision making** in your field of interest.

What to Submit

To complete this activity, you must submit the following:

Python Code

Download your Jupyter Notebook as an HTML file. Review the file to ensure the code and the final table output are displayed. Submit the HTML file.

Reflection

Submit your reflection as a 1-page Microsoft Word document with double-spacing, 12-point Times New Roman font, and one-inch margins.

AI Usage

If you use gen AI tools to support your work on this assignment, be sure to follow [these AI usage guidelines](). You must acknowledge your use of these tools in your work. Guidelines on how to cite AI tools can be found in [this Shapiro Library guide]().

Module Two Activity Rubric

Criteria	Exceeds Expectations (100%)	Meets Expectations (90%)	Partially Meets Expectations (70%)	Does Not Meet Expectations (0%)	Value
Part One: Structure	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Examines the **structure** of the provided data set using Python code	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	15
Part One: Python Functions	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Cleans the raw data using Python functions to remove and standardize messy entries	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	20

Criteria	Exceeds Expectations (100%)	Meets Expectations (90%)	Partially Meets Expectations (70%)	Does Not Meet Expectations (0%)	Value
Part One: Tabular Format	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Converts the cleaned data into a tabular format	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	15
Part One: Validate Table	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Validates the table for consistent column names, correct data types, and missing values	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	20
Part Two: Support Analysis and Decision Making	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Explains how transforming raw data into table format supports analysis and decision making in the field of interest	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	25
Clear Communication	Exceeds expectations with an intentional use of language that promotes a thorough understanding	Consistently and effectively communicates in an organized way to a specific audience	Shows progress toward meeting expectations, but communication is inconsistent or ineffective in a way that negatively impacts understanding	Shows no evidence of consistent, effective, or organized communication	5
				Total:	100%